

The Distress Signal Monitor

An early-warning layer for neighborhood mental health — reading demand before it becomes a 911 call

WHAT IT IS

A ZIP-code-level **weekly distress index** built by mining public social media posts (Twitter/X, Reddit community boards, Nextdoor) with NLP models trained on mental-health crisis language.

The system aggregates distress-keyword hit rates and sentiment scores, anchors them against CDC BRFSS baseline mental-health prevalence, and flags ZIP codes where the index exceeds a threshold.

Alerts route to **crisis coordinators and community health workers** so outreach can be dispatched three weeks earlier than a 211 spike would surface the same surge.

The product is **purely signal-side**: it detects demand, not response. It is the earliest layer of a three-layer warning pipeline.

HOW IT COMPARES

211 Crisis Lines

+3 weeks late

Reactive — only registers demand after a person calls.

CDC BRFSS Surveys

1-year lag

Annual, state-level. Too coarse for neighborhood dispatch.

Emergency Dispatch Logs

Crisis stage

Captures the crisis after police or EMS are already involved.

Distress Signal Monitor

Day 3 of surge

ZIP-level, weekly, pre-call. Reads public posts as a demand signal.

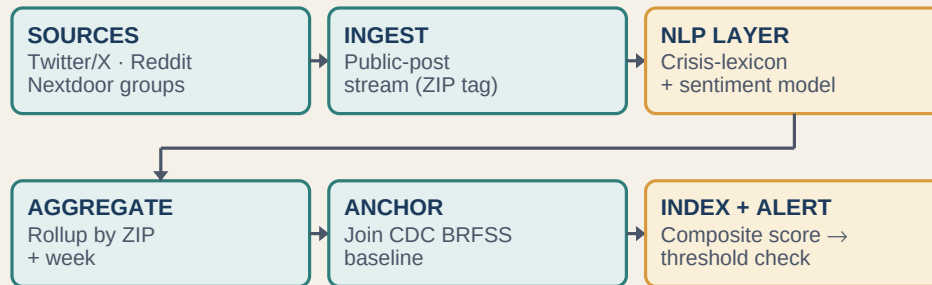
KEY DIFFERENTIATOR

The warning is already there — written publicly by people who need help. We just read it first.

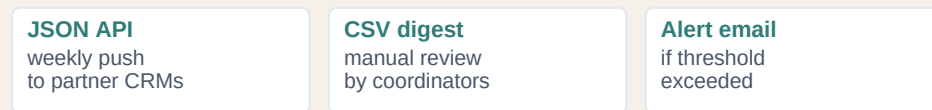
Blueprint — How Information Flows and Surfaces

A data-pipeline blueprint paired with the coordinator's dashboard view

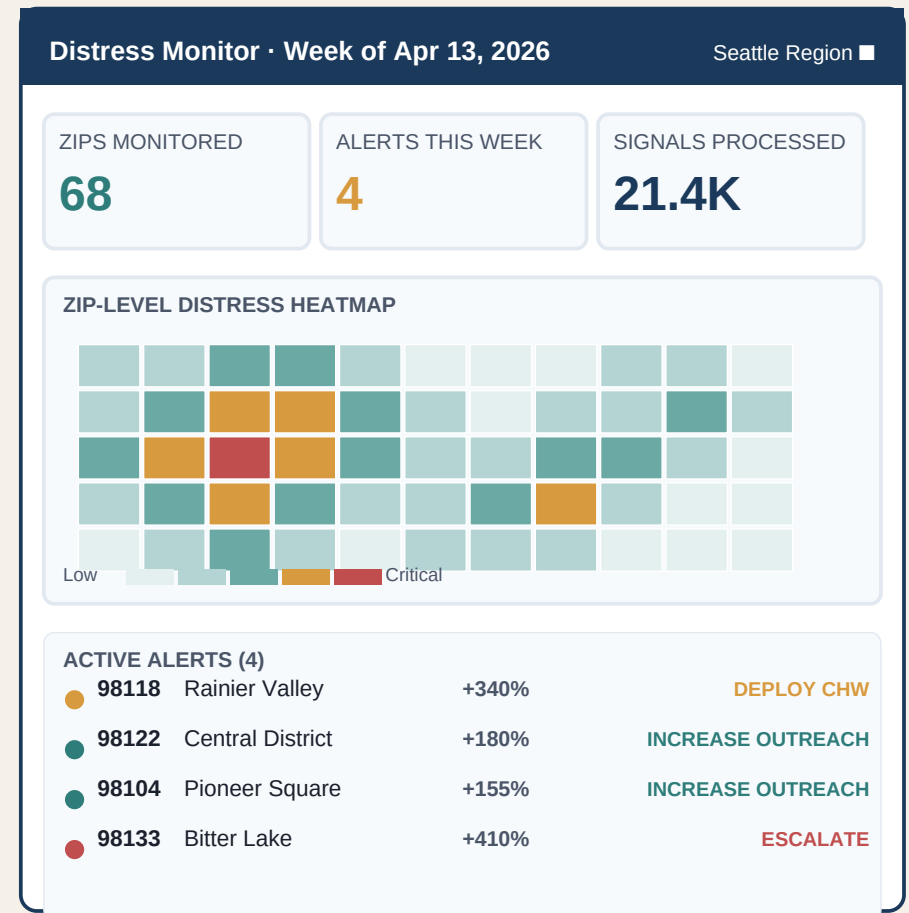
INFORMATION FLOW BLUEPRINT



DELIVERY



COORDINATOR DASHBOARD (UI WIREFRAME)



Main Concepts & Metadata Scheme

The labeling system — shared terms that let every downstream system understand a distress signal the same way

TAXONOMY

DistressSignalRecord

■ Geography

- zip_code (5-digit USPS)
- region_name (human label)

■ Temporal

- week_start_date (ISO 8601)
- week_end_date (ISO 8601)

■ SignalMetrics

- total_posts_analyzed
- distress_keyword_hit_rate (0-1)
- sentiment_distress_score (0-1)
- week_over_week_change_pct

■ BaselineAnchor

- brfss_poor_mental_health_days

■ Index & Action

- composite_distress_index (0-100)
- alert_threshold_exceeded (bool)
- recommended_action (enum)

■ Provenance

- data_source (string[])

VOCAB: RECOMMENDED_ACTION

Monitor

No intervention; continue weekly tracking

Increase Outreach

Push extra community messaging into the ZIP

Deploy Community Health Worker

Dispatch a worker for on-the-ground contact

Escalate to Crisis Coordinator

Hand off to county-level coordinator for surge response

SUPPORTING VOCABULARIES

data_source (multi-valued)

twitter_x

reddit_community

nextdoor

public_blog

threshold_tier (derived)

● green

● yellow

● amber

● red

Portability keys

- ZIP (geo join)
- ISO 8601 (time join)
- 0–1 normalized scores

LABELING PRINCIPLE

Every field is named in plain, non-jargon terms so a 211 operator, a city analyst, and a nonprofit caseworker can all read the same record and act on it.

Hosting · Information Structure · Technology Stack

A pragmatic v1 stack — public-data-first, cloud-hosted, open-format outputs

LAYERED ARCHITECTURE (V1)

1. INGESTION

Public APIs + scrapers · ZIP-tagged post stream

• Twitter/X API v2

• Nextdoor public posts

• Reddit API (PRAW)

• Python + AWS Lambda (scheduled)

2. PROCESSING

NLP classification + sentiment + ZIP aggregation

• spaCy + HuggingFace transformers

• Pandas for weekly rollups

• Custom crisis-lexicon model

• AWS ECS (containerized batch)

3. STORAGE

Warehouse for signals + anchors + derived indices

• PostgreSQL (core records)

• CDC BRFSS loaded as reference table

• S3 (raw post archive, encrypted)

• Partitioned by ZIP + week

4. DELIVERY

JSON API + CSV digest + alert email

• FastAPI (public JSON endpoints)

• SendGrid alert emails

• Scheduled CSV export to partner SFTP

• Hosted on AWS (us-west-2, Seattle)

PRIVACY

Only public posts. Aggregated to ZIP. No individual identifiers stored.

OPEN STANDARDS

JSON + CSV outputs. ZIP & ISO 8601 as portability keys.

GOVERNANCE

Weekly review board. Human-in-the-loop before any alert escalation.